

Task 2

What is the difference between (int, double, float, decimal)?!

1 int (Integer)

- What it stores: Whole numbers (no decimal point)

```
x = 10y = -5z = 1000000
```

- ✓ Exact precision
- ✓ No size limit (Python integers can grow very large)
- ✗ Cannot store fractional

numbers Example:

```
a = 5b = 2  
print(a / b)
```

```
# 2.5 (result becomes float)
```

2 float (Floating Point Number)

- What it stores: Numbers with decimal points

```
x = 3.14y = -0.5
```

- Internally uses 64-bit double precision
- Approximate (not perfectly precise!)
- ⚠ Precision problem
- example:

```
print(0.1 + 0.2)
```

- Output:

```
0.30000000000000004
```

- This happens because floating-point numbers are stored in binary format.

✓ Fast

✓ Good for scientific calculations

✗ Not perfectly precise for financial calculations

3 decimal (High Precision Decimal)

- From the built-in decimal module:
- from decimal import Decimal

```
x = Decimal("0.1")  
y = Decimal("0.2")  
y = Decimal("0.2")
```

Output:

```
0.3
```

- ✓ Exact decimal precision
- ✓ Best for money & financial calculations
- ✗ Slower than float
- ✗ Must import module

4 double in Python?

- There is no separate double type.
- In languages like: C, C++, Java
 - float = 32-bit
 - double = 64-bit
- But in Python:
 - float = 64-bit double precision
- So Python's float is already equivalent to double.